Moffat Bay Marina

Website Design Document

Introduction

This document will provide technical implementation details for a website that is to be created for Moffat Bay Marina. Within this document you will find a Terminology section for all stakeholders to reference.

Purpose

Moffat Bay Marina currently offers long-term slip reservations that can be booked in person, over the phone and via email. Their office manager tracks reservations with a spreadsheet that they manually update. Moffat Bay Marina does not currently have a website, so customers must call or email to request information regarding the size of slips and reservation dates available. This process requires careful coordination and manual data entry, which can be time-consuming.

Moffat Bay Marina would like a website that can provide details about the marina and slip availability to any visitors. They would like their website to manage reservations and user accounts for their customers. It is important to note that, while a customer account needs to be created to utilize the online booking system, the website itself should be viewable to any visitors regardless of customer account creation. Additionally, the reservation system will not process payments for customers and will not require payment to confirm a reservation.

Terminology

Marina/Business terms

Slip - a designated area in a marina where a boat can "park" (Discover Boating, n.d.)

Harbormaster - a person who oversees harbor or marina traffic and manages anchorages (Discover Boating, n.d.)

VHF - stands for "Very High Frequency" and is a bandwidth designation typically used by marine radios (Discover Boating, n.d.)

Long-Term Reservation - reservations that last for a season (several months) to an entire year, as opposed to short-term reservations, which are measured in days, weeks or months

Technical/System terms

Authentication - "in web security, it is the process of verifying the claimed identity of some entity, such as a user. This then makes it possible to decide whether to grant the user the access that they are requesting, such as being signed into a particular account." (MDN contributors, 2025a)

User - in most cases, this will refer to the customer, but can refer to any person who is using a system (in this case, the website)

User account – in most cases this will be refer to a customer account, but Moffat Bay Marina staff will have user accounts, as well, with varying permissions.

CustomerID - a unique identifier for each customer to be stored in the Moffat Bay Marina database

ReservationID - a unique identifier for each reservation made to be stored in the Moffat Bay Marina database

Session - a window of time in which a user accesses a website

Landing Page - the main page of a website through which a user can access other sections of the website. The landing page will typically contain basic information about the business and navigation elements, like a menu

Form / Form Field - in HTML, a form contains elements, like fields, that can receive user input (ex. Email field) or user interactions (ex. Clicking the "submit" button after filling out an email field) (MDN contributors, 2025c)

Database - a system that collects and stores organized data. Databases make searching and structuring information easier and more efficient. (Erickson, 2024)

Open Source - publicly available code/designs

RDBMS - stands for Relational Database Management System (Erickson, 2024)

MySQL - an open-source RDBMS that is commonly used to manage databases for companies (Erickson, 2024)

Regular Expression (Regex) - a pattern-matching mechanism that can be used to check whether text conforms to a specific format. For example, a phone number is 10 digits and is typically formatted XXX-XXX-XXXX

ERD - stands for "Entity-Relationship Diagram"; this is a visual flowchart typically used in database design that illustrates how entities (ex. Customers, slip reservations) are related within a system

HTML - stands for "HyperText Markup Language"; this is the standard language that is used to create website content (MDN contributors, 2025c)

CSS - stands for "Cascading Style Sheets"; used for the visual organization of HTML (MDN contributors, 2025c)

JSP - stands for "JavaServer Pages"; a technology that is used to create web pages that can display changing information (dynamic) from a Java-based web application. For example, if a customer visits the Moffat Bay Marina website and logs into their account, the banner on the webpage might display their name in the upper right corner to indicate they are logged in. (MDN contributors, 2025b)

PII - stands for "Personally Identifiable Information"; this can be any data that can be used to identify a specific person. For example, a person's social security number would be considered PII.

User Personas

User 1: Angela

Gender: Female
Age: 42
Occupation: Elementary Teacher
Education: College/Undergraduate
Marital Status: Married
Location: Local, lives in town year-round

Figure 1

Note. AI-generated image created with Google Gemini (Google, 2026).

Background

Angela and her family have had a 40 ft boat for six years now, and they book a long-term slip every season. She's got a full-time job and two kids with their own schedules, so she doesn't have time to sit on the phone dealing with reservations. She just wants something quick that works.

Angela's Needs

- An option to see what slips are open without calling the business
- Some kind of confirmation she can send to her husband once it's booked
- An account that remembers her info so she's not typing the same stuff in every season

Angela's User Stories

1. As a returning customer, I want to check slip availability online so I can plan my season without calling the office.
2. As a busy parent, I want my account and boat info saved after I register so I don't have to re-enter them every time.
3. As a slip holder, I want confirmation that I can save or forward the reservation details to share with my family.

User 2: Jesse

Gender: Male
Age: 47
Occupation: Harbormaster
Education: Community College/Vocational School
Marital Status: Married
Location: Local, lives in town year-round

Figure 2

Note. From *Harbor and Marina Staff Directory, by City of Monterey (n.d.).*

Background

This is Jesse, he is a staff member at the Moffat Bay Marina. Jesse's goal is to keep long-term slip reservations organized and make sure customers are assigned to slips that fit their boats.

Jesse's Needs

- Accurate reservations
- Slip availability
- Efficient access to registered customer information
- An easy way to confirm reservations.

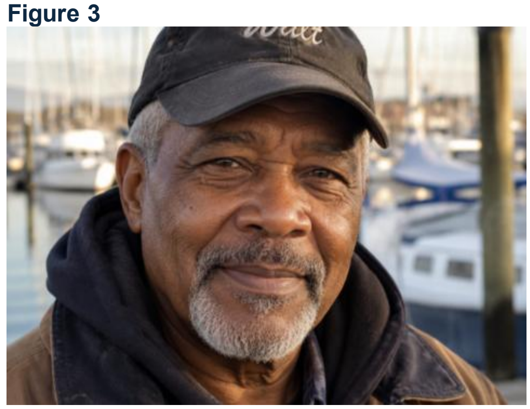
Jesse's User Stories

1. As a marina staff member, I want to view current slip availability so that I can know which long-term slips are still available.
2. As a marina staff member, I want customers to be matched with the right slip size based on the length of their boat.
3. As a marina staff member, I want to see confirmed reservations so that I can track available slips and avoid double bookings.

User 3: Walt

Gender: Male
Age: 68
Occupation: Retired Electrician
Education: High School
Marital Status: Widowed
Location: Winter Visitor

Figure 3



Note. AI-generated image created with ChatGPT (OpenAI, 2026).

Background

Walt is a 68-year-old retired electrician and widower who escapes the snow each winter at Moffatt Bay Marina Lodge. From November through February he stays at the lodge for three to four months, keeps his boat in a dock slip, and spends most days out fishing. Since losing his wife, Walt values reliability, simplicity, and a sense of community.

Walt's Needs

- A reserved slip for his entire 3-4 month winter stay
- Easy, bundled booking that hapirs his lodge room with his boat slip
- Peace of mind that his boat is secure between fishing trips

Walt's User Stories

1. As a winter dock guest, I want to reserve a dock slip for a full 3–4-month block in one transaction, so I don't have to re-book or risk losing my spot month to month.
2. As a guest who is older, I would prefer to select a dock slip that is closer to the shore, so I don't have to haul my equipment as far when I take it to and from the lodge.
3. As a long-term slip holder, I would like clear, upfront pricing for a multi-month stay, including any seasonal or long-term discounts, so that I can budget for the winter without surprise fees.

Moffat Bay Marina - Group Task List & Work Estimations

Each page is broken into Front End, Back End, and Database needs. Estimated hours are approximate - see citations below.

Task	Description	Estimated Hours	Due Date
MODULE 1 - Project Setup & Requirements Gathering (due Aug 16) - DONE			
Requirements Gathering	Draft data requirements, user personas, and user stories for the TDD	6 hrs	Aug 16 (Done)
Project Setup	Confirm Marina project, set up GitHub repo, shared MySQL environment, comms channel	4 hrs	Aug 16 (Done)
MODULE 2 - Prototype (due Aug 23)			
Page Prototypes	Functional prototype for all 9 pages (Landing, About Us, Contact Us, Wait List Lookup, Registration, Login, Slip Reservation, Reservation Summary, Look Up Reservation)	9 hrs	Aug 23
Landing Page Mockup	Realistic mockup (color, layout, font) of the Landing page	3 hrs	Aug 23
MODULE 3 - ERD (due Aug 30) - DATABASE			
Entity Relationship Diagram	Design the ERD covering customers, slips, reservations, and wait list, and how they relate	6 hrs	Aug 30

Task	Description	Estimated Hours	Due Date
MODULE 4 - Database Development, Sprint 1 (due Aug 30) - DATABASE			
Database Schema	Build the MySQL schema from the ERD (customers, slips, reservations, wait list tables)	6 hrs	Aug 30
Slip Inventory	Load slip inventory by size category (26 ft, 40 ft, 50 ft) based on the marina map	2 hrs	Aug 30
Shared Data Access Functions	Build shared functions so every page has a consistent way to read/write the database	7 hrs	Aug 30
Password Hashing	Research and implement password hashing approach (cited source)	4 hrs	Aug 30
MODULE 5 - Web Dev 1, Sprint 2 Begins (due Sep 6)			
Landing Page - Front End	HTML/CSS build	3 hrs	Sep 6
Landing Page - Back End	Any dynamic content	2 hrs	Sep 6
Landing Page - Database	*None - static content*	-	*Sep 6*
Login Page - Front End	Username/password form	2 hrs	Sep 6

Task	Description	Estimated Hours	Due Date
Login Page - Back End	Check credentials, hash check, start session	4 hrs	Sep 6
Login Page - Database	*Reads customers table (username, password hash)*	-	*Sep 6*
Registration Page - Front End	Registration form	2 hrs	Sep 6
Registration Page - Back End	Create customer record, validate email format, hash password	4 hrs	Sep 6
Registration Page - Database	*Writes to customers table (new row, unique CustomerID, hashed password)*	-	*Sep 6*
Module 5 Test Plan	6 functional tests (2 per page) using TestPlanTemplate.docx, run and document with screenshots	3 hrs	Sep 6
MODULES 6 & 7 - Web Dev 2 & 3, Sprint 2 Continues (due Sep 13)			
Slip Reservation Page - Front End	Reservation form	3 hrs	Sep 13
Slip Reservation Page - Back End	Boat-length-to-slip-size matching, availability check	5 hrs	Sep 13
Slip Reservation Page - Database	*Reads slips (availability by size), writes reservations*	-	*Sep 13*

Task	Description	Estimated Hours	Due Date
About Us Page - Front End	HTML/CSS build	2 hrs	Sep 13
About Us Page - Back End	Any dynamic content	1 hr	Sep 13
About Us Page - Database	*None - static content*	-	*Sep 13*
Reservation Summary Page - Front End	Summary display	2 hrs	Sep 13
Reservation Summary Page - Back End	Pull reservation + slip details for display	3 hrs	Sep 13
Reservation Summary Page - Database	*Reads reservations, slips, customers*	-	*Sep 13*
Module 7 Test Plan	4 functional tests (2 per page)	2 hrs	Sep 13
Scrum 2 Discussion	Initial post plus 3 peer responses (per person)	1 hr/person	Sep 13
MODULE 8 - Web Dev 4, Sprint 3 Begins (due Sep 20)			
Contact Us Page - Front End	HTML/CSS build	2 hrs	Sep 20

Task	Description	Estimated Hours	Due Date
Contact Us Page - Back End	Any dynamic content	1 hr	Sep 20
Contact Us Page - Database	*None - static content (unless a stored contact form is added later)*	-	*Sep 20*
Look Up Reservation Page - Front End	Lookup form	2 hrs	Sep 20
Look Up Reservation Page - Back End	Query reservation by ID/customer	3 hrs	Sep 20
Look Up Reservation Page - Database	*Reads reservations*	-	*Sep 20*
Module 8 Test Plan	4 functional tests (2 per page)	2 hrs	Sep 20
Scrum 3 Discussion	Initial post plus 3 peer responses (per person)	1 hr/person	Sep 20
MODULE 9 - Web Dev 5, Sprint 3 Wraps Up (due Sep 27)			
Wait List Lookup Page - Front End	Wait list display	2 hrs	Sep 27
Wait List Lookup Page - Back End	Wait-list logic when a slip size is full	4 hrs	Sep 27

Task	Description	Estimated Hours	Due Date
Wait List Lookup Page - Database	*Reads/writes wait_list, reads slips*	-	*Sep 27*
Module 9 Test Plan	2 functional tests	1 hr	Sep 27
MODULE 10 - QA & Peer Review (due Oct 4)			
QA Presentation	Team intro, prototypes, ERD, 5+ tests and results, lessons learned	4 hrs	Oct 4
Screen Capture Walkthrough	Narrated recording walking through every page of the site	3 hrs	Oct 4
MODULE 11 - QA Testing & Delivery (due Oct 11)			
Peer Critiques	2 critiques of other teams, 150+ words each, 2 strengths/2 improvements	2 hrs/person	Oct 11
Peer Feedback Fixes	Fix issues raised in peer feedback across pages/database	5-8 hrs	Oct 11
Changes Write-Up	Document the changes made based on feedback	2 hrs	Oct 11
Final Submission	Zip and submit the final project	1 hr	Oct 11

How These Estimates Were Generated

We started with two numbers we found while researching typical web project timelines: about 40 hours for planning and design, and roughly 5 hours to build each page. For the login page, we used a starting point of 1 to 3 hours for a basic username and password check. From there, we adjusted each task up or down by hand. A page like Slip Reservation, which has to match a boat's length to an available slip, took more time than something like the About Us page. It's worth knowing that both of these numbers came from articles written to help price out client projects, not from a formal study of how long development actually takes. So, think of them as a solid starting point, not a guarantee.

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